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## 2.1 Introduction

Almost all human activities are based on light. Natural light is obtained from the sun. Artificial light plays an important role in our everyday life. In places where natural light is not available, artificial light is obtained by electric lamps. Lighting plays an important role because of its belief, consistency, simple control and low price. The electrical lighting is mainly used for domestic purpose, decorative purpose, advertising, traffic- control, medical field and for street lighting also.

When light falls on a surface, it becomes visible and the phenomenon is called as illumination. It is denoted by E and is measured in lumen per square meter.

## 2.2 Important terms in illumination

### 2.2.1 Plane angle

The angle subtended at a point by two converging lines lying in the same plane is called plane angle, and is measured in radians. It is shown in figure 2.1. It is equal to the ratio of the length of the arc to its radius.

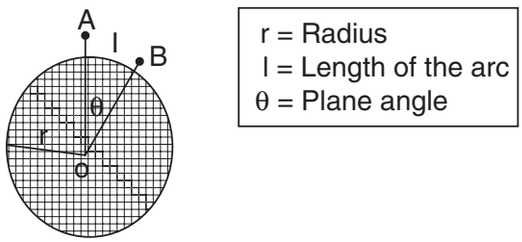



**Fig. 2.1 Plane angle**

$$\theta = \frac{Arc}{\text{radius}} \text{radian}$$

$$\theta = \frac{l}{r} \text{radians}$$

### 2.2.2 Solid angle

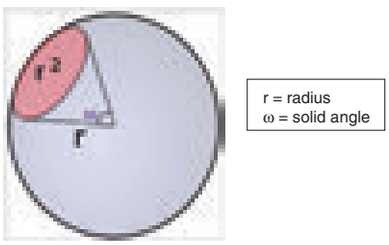



**Fig. 2.2 Solid angle**

Solid angle is measured in steradians. Solid angle is the ratio of area of the surface to the square of radius of sphere. It is shown in figure 2.2.

$$\omega = \frac{Area\ of\ surface}{\text{Square of radius}} \text{steradians}$$

$$\omega = \frac{A}{r^2} \text{steradians}$$




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